Shivani Swaminathan

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science / Master of Science in Aerospace Engineering — *B.S. Fall 2027 | M.S. Fall 2028*
GPA: 3.8/4.0 | Dean's List | Relevant Coursework: Aerodynamics, Structural Analysis, Fluids, Thermodynamics, Mechanics of Materials

Research

Ben T. Zinn Combustion Laboratory — Atlanta, GA
Undergraduate Research Assistant — *May 2026 – Present*
- Improved combustor-model fidelity by 18% across 8 steady and transient ANSYS Fluent cases by refining 19M-cell meshes and configuring reacting-flow physics for higher-confidence combustion predictions.
- Validated 3 combustor cases by analyzing temperature, pressure, velocity, and species results with ANSYS engineers, leading to 4 model and setup corrections and a more reliable simulation baseline.

GT Aerospace Systems Design Laboratory (ASDL) — Atlanta, GA
Undergraduate Research Assistant – Systems Modeling — *August 2025 – Present*
- Developed an executable SysML mobility subsystem linking torque, traction, speed, power, and climb constraints, then evaluated 30+ parametric cases to establish feasible performance limits.
- Built a traceable SysML model linking mission requirements, UAV architecture, operating behavior, and performance equations; evaluated 3 delivery scenarios against 10 km range, 40-minute battery, and 2 kg payload requirements.

GT Aerospace Systems Design Laboratory (ASDL) — Atlanta, GA
Undergraduate Research Assistant – AI/ML — *June 2026 – Present*
- Developed a Python AI agent that expanded a lunar-base simulation to 12+ autonomous construction tasks, automating landing-pad sequencing, equipment allocation, and task dependencies for repeatable mission-plan evaluation.
- Integrated 4 lunar infrastructure models across 4 mission scenarios by defining dependencies among landing-pad construction, in-situ resource utilization (ISRU) mining, and power operations, enabling end-to-end surface construction simulation.

UGA Department of Physics and Astronomy — Athens, GA
Undergraduate Research Assistant — *August 2024 – May 2025*
- Analyzed 200+ stellar-flare light curves in Python and modeled activity decay across 3 spectral classes in MATLAB, reducing processing time by 36% and earning an undergraduate research award.

Experience

GT HyTech Racing — Atlanta, GA
Aerodynamics Engineer — *August 2025 – Present*
- Validated carbon-fiber wing assemblies across 6+ ride-height and angle-of-attack cases in ANSYS Fluent, identifying up to 18% downforce and 11% drag variation to select competition wing ride height and angle of attack.
- Reduced analysis turnaround by 36% by automating ANSYS Fluent post-processing in Python, enabling same-day comparison of 8+ aerodynamic designs for faster design decisions.

Handshake AI — Remote
CAD Specialist – OpenAI-Sponsored AI Evaluation Project — *March 2026 – Present*
- Developed 25+ CAD benchmark tasks and weighted rubrics in SolidWorks, Siemens NX, and AutoCAD to evaluate AI-generated parts, assemblies, and drawings against measurable geometry, dimensioning, and deliverable criteria.
- Evaluated 40+ developer-created CAD benchmarks and identified 60+ dimensional, manufacturability, feature-quality, and deliverable issues, improving task clarity and scoring consistency before AI model testing.

Projects

Wind Turbine Blade Fluid–Structure Interaction Analysis — *July 2026*
- Integrated CFD and FEA across 6 operating conditions, linking aerodynamic loads to blade stress and deformation while validating root radial force within 0.1% of hand calculations.

Unified Aerodynamic Analysis Tool — *August 2026*
- Developed a MATLAB aerodynamic solver combining Prandtl–Glauert-corrected thin-airfoil and shock-expansion theory, then searched 45,000+ configurations to optimize a double-wedge airfoil across subsonic and supersonic regimes.

CubeSat Frame & Attitude Control Design — *January 2025 – March 2025*
- Reduced CubeSat frame mass by 10% by iterating structural layout against 5g stiffness, safety-factor, and buckling constraints in ANSYS; developed a MATLAB/Simulink PD controller achieving 3-axis stabilization.

Skills

Simulation: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), Simscape Multibody
3D Modeling: SolidWorks (CSWA), CATIA, Inventor, AutoCAD, Fusion 360, Siemens NX
Systems Engineering: MagicDraw/Cameo (SysML), ModelCenter, Simulink
Programming: MATLAB (Certified MATLAB Associate), Python, Git